

Shihao Cheng (程世豪)

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I am a **Master's student** advised by Prof. Zhigang Tu at LIESMARS, **Wuhan University**, China. Expected graduation: June 2027. Prior to this, I obtained my B.S. degree from **Harbin Institute of Technology** in 2024, ranking **first** in my class. Currently, I am a Research Intern at **Tencent Hunyuan**, focusing on agentic streaming video generation and world modeling via closed-loop feedback. Before this, I interned at TeleAI, working on **streaming audio-video joint generation**. My research interests lie in multimodal understanding and generation, with a particular focus on solving complex alignment and interaction challenges in generative models. I have published or submitted works to **CVPR (Highlight)**, **ECCV**, and **T-CSVT**.

EDUCATION

Wuhan University

M.S. Student. Communication and Information Systems

Wuhan, Hubei
Sep. 2024 – Now

Harbin Institute of Technology

Bachelor of Information Engineering

Harbin, Heilongjiang
Sep. 2020 – Jun. 2024

- Avg Score: 92.3/100, Rank: 1/29.
- **National Scholarship 2023. ¥8,000 RMB, Top 1%.** Provincial Outstanding Graduate, 2024, Top 1%.

SELECTED PUBLICATIONS

Unison: Harmonizing Motion, Speech, and Sound for Human-Centric Audio-Video Generation

European Conference on Computer Vision (ECCV 2026) | First Author | (Under Review)

- Resolved speech-SFX interference and motion-audio desynchronization in audio-video generation.

InteractiveAvatar: Real-Time Streaming Video Generation for Consistent and Intent-Aware Avatars

European Conference on Computer Vision (ECCV 2026) | (Under Review)

- Resolved the understanding interaction and long-term semantic drift in infinite-length video generation.

GeoMMBench and GeoMMAgent: Toward Expert-Level Multimodal Intelligence in Geoscience and Remote Sensing

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026, Highlight) | Co-First Author

- Resolved the lack of expert-level evaluation benchmarks for interdisciplinary and multi-sensor scenarios, and built a multi-agent architecture to address this issue.

OwlSight: A Robust Illumination Adaptation Framework for Dark Video Human Action Recognition

IEEE Transactions on Circuits and Systems for Video Technology, 2025 (T-CSVT, IF: 11.1) | First Author

- Resolved inefficient brightness utilization in low-light enhancement and action recognition end-to-end training.

EXPERIENCE

Research Intern, Tencent Hunyuan

Apr. 2026 – Now

- I participated in the research of agentic video world modeling to synergize reasoning and generation, focusing on streaming audio-visual generation with semantic-temporal alignment via Hierarchical World State Memory.

Research Intern, TeleAI

Aug. 2025 – Apr. 2026

- Advised by: Dr. Shansong Liu
- Participated in research on audio-video synchronous and streaming video generation models, contributing to the development of a real-time, human-centric video generation system with audio-visual synchronization, which has been successfully launched on **Telestudio**.

Research Project (SenseTime): Audio-Visual Semantic and Temporal Alignment in MLLMs

Mar. 2025 – Jun. 2025

- Advised by: Dr. Jiahao Wang
- Investigated Multimodal Large Language Models (MLLMs) for joint audio-visual understanding, focusing on resolving complex semantic and temporal alignment challenges. Designed and optimized multimodal fusion modules to enhance cross-modal representation learning and temporal synchronization.